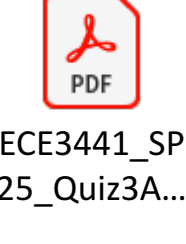


Quiz 3

Sunday, April 27, 2025 9:42 PM



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ECE 3441 Spring 2025

Quiz 3

Version A

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 50 minutes.

Please do not open booklet unless instructed to do so, or substantial penalty will result.

SOLUTIONS

Last Name, First Name Peoplesoft ID, Date (YYYY/MM/DD)

Question	Points	Maximum
1		25
2		30
3		30
4		30

Total: _____/(100 possible points + 10 bonus points)

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Question #1: *Yab 6/7*

Using D flip-flops, design a 3-bit counter that counts in the following sequence. Provide the state table, transition table, and next-state equations. Show all of your work clearly.

Hint: this is a six-state counter, which means the two unused states are don't-cares (Xs).

① Create Q/Q+ table

Present	Next
001	110
110	101
101	100
100	000
000	010
010	001

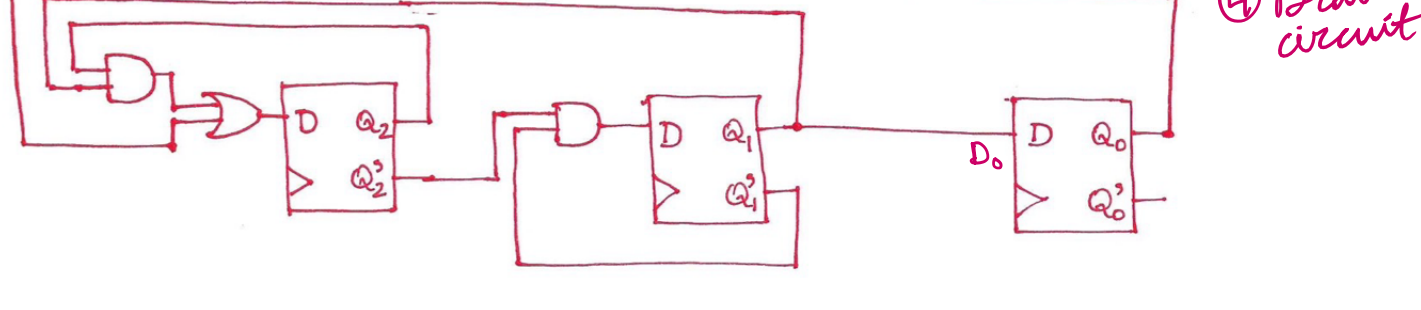
② Transition table

$Q_2 Q_1 Q_0$	$Q_2^+ Q_1^+ Q_0^+$
0 0 0	0 1 0
0 0 1	1 1 0
0 1 0	0 0 1
0 1 1	X X X
1 0 0	0 0 0
1 0 1	1 0 0
1 1 0	1 0 1
1 1 1	X X X

③ Kmaps for equations

④ Draw circuit

$D_2 = Q_0 + Q_2 Q_1$; $D_1 = Q_2^2 Q_1^2$; $D_0 = Q_1$ ← *simplified from Kmaps*



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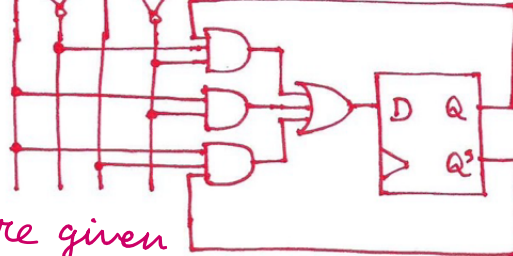
Question #2:

Show how a D flip-flop can be converted to a J-K flip-flop by adding external gates or a MUX.

Hint: Draw the TT for a JK FF, then fill in the D input based on Q/Q+ for the JK FF to simulate it.

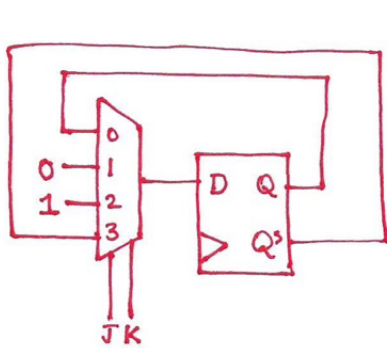
① Write truth table for flip-flop you're trying to get to

JK	Q+	D
0 0	0	0
0 1	0	0
1 0	1	1
1 1	Q	Q



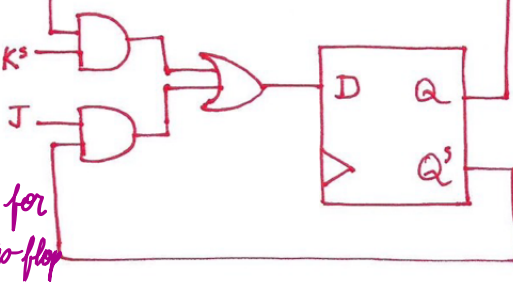
③ Make table for flip-flop you're given

JK	Q	Q+
0 0	0	0
0 1	0	1
1 0	0	1
1 1	0	0
1 1	1	1
1 1	1	0



③ Do Kmaps to find equation

JK	Q	Q+
0 0	0	0
0 1	0	1
1 0	0	1
1 1	0	0
1 1	1	1
1 1	1	0



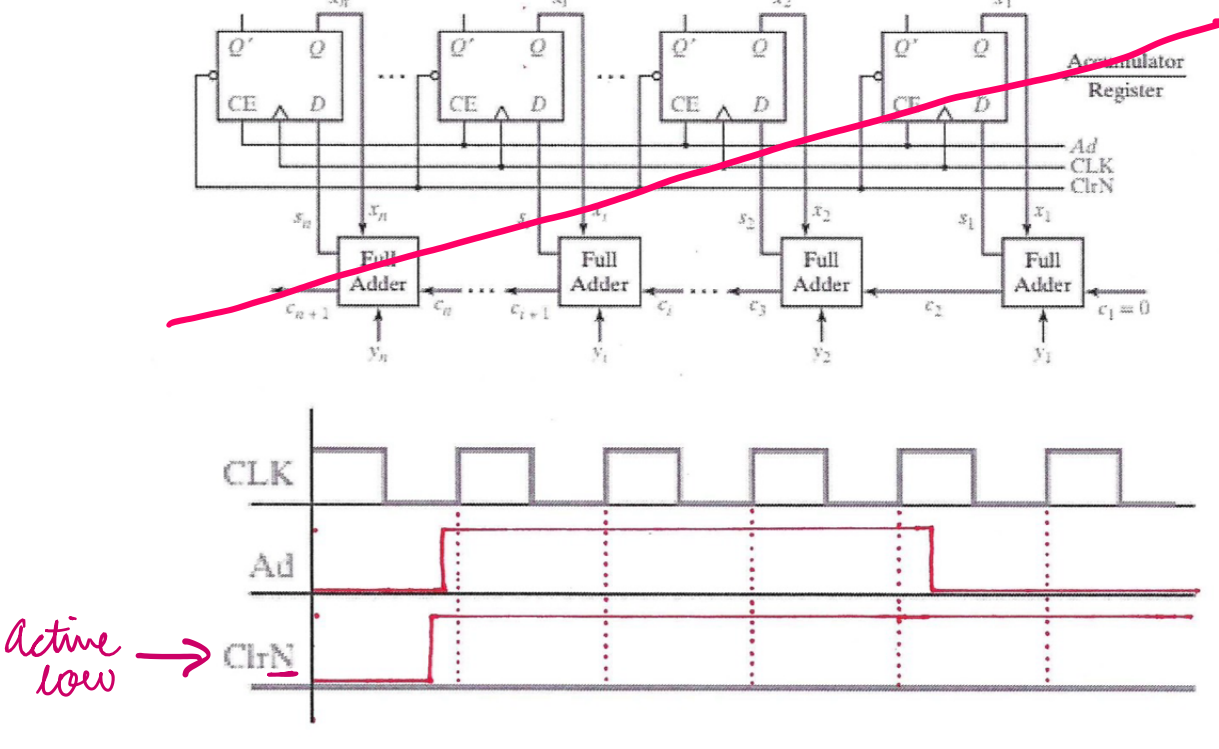
$\Rightarrow D = Q^+ = JQ^0 + K^0Q$ simply for JK $\Rightarrow$ D flip-flop

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Question #3:

Consider a 4-bit adder with an accumulator shown below. Suppose the X register contains a number from a previous calculation. We do not want this number. Instead, we want X to hold the value $4 \times Y$. ($X = X_3X_2X_1X_0$ and $Y = Y_3Y_2Y_1Y_0$). On the timing diagram, give values for Ad and ClnN so that we will have $4 \times Y$ held in the accumulator.

Hint: First, you need to clear X, then add Y four times, on five successive clock cycles.



Ad needs to be high for exactly 4 rising edges of the clock

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Question #4: *when clock "falls"*

Suppose you are given a falling-edge triggered "Y-Z flip-flop" which behaves as follows:

If YZ = 00, then Q is toggled

If YZ = 01, then Q is set to 0

If YZ = 10, then Q is set to 1

If YZ = 11, then Q is unaffected

Complete the given timing diagram by assuming negligible propagation delay.

